Algorithms

Lecture #37

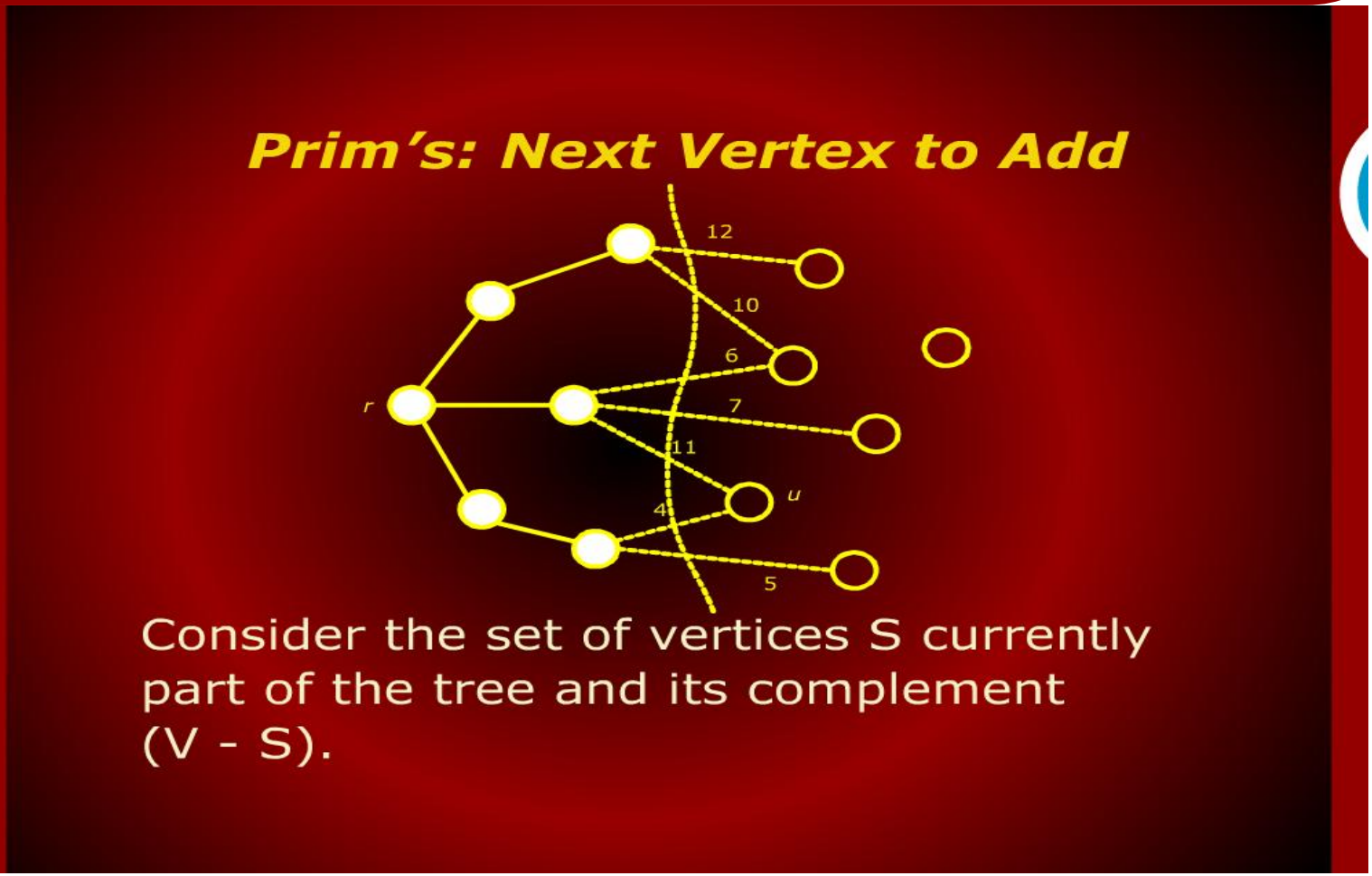
Syed Hamed Raza

Prim's Algorithm

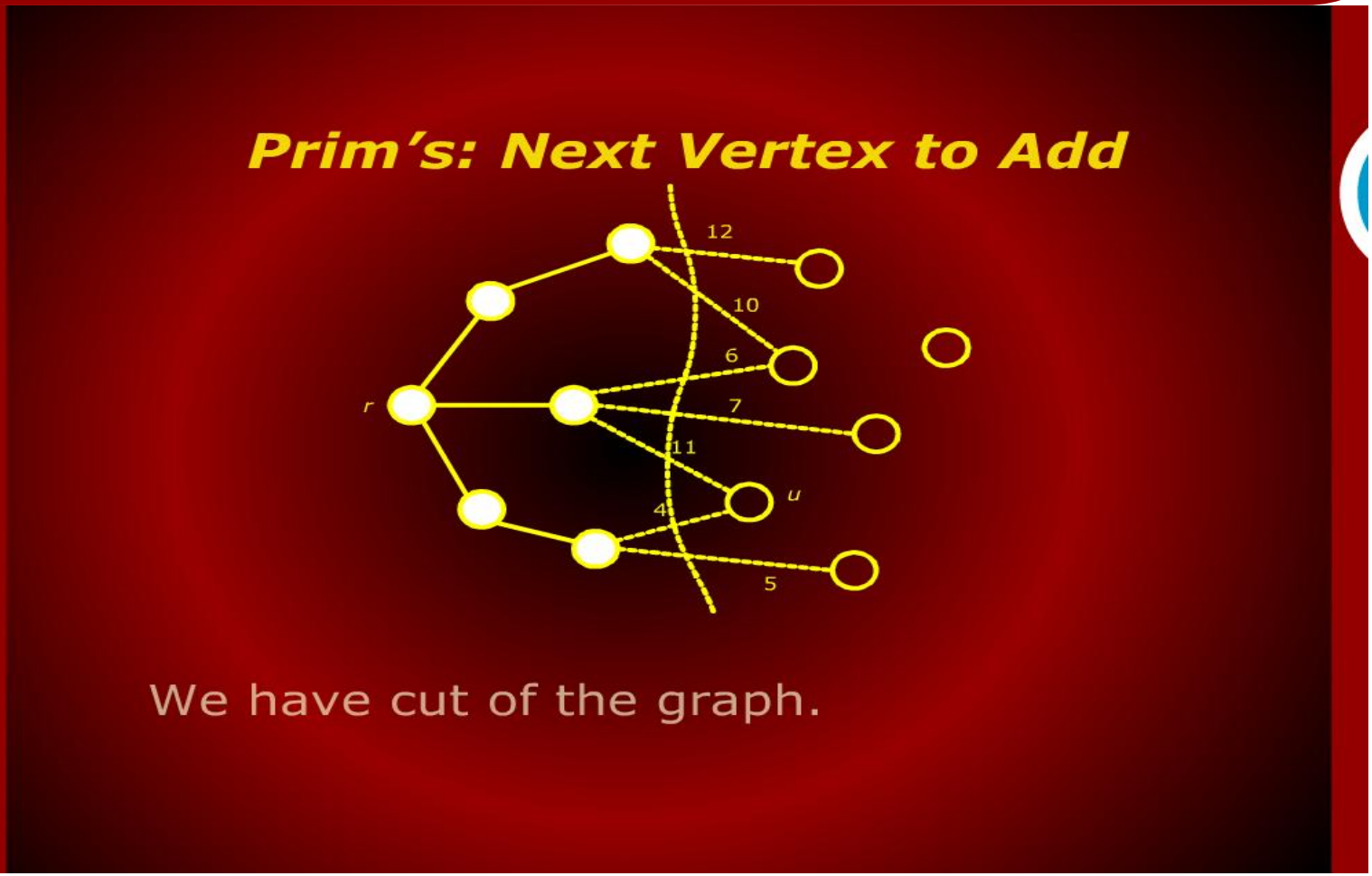
Prim's Algorithm

- Prim's algorithm builds the MST by adding leaves one at a time to the current tree.
- We start with a root vertex r; it can be any vertex.
- At any time, the subset of edges A forms a single tree (in Kruskal's, it formed a forest).

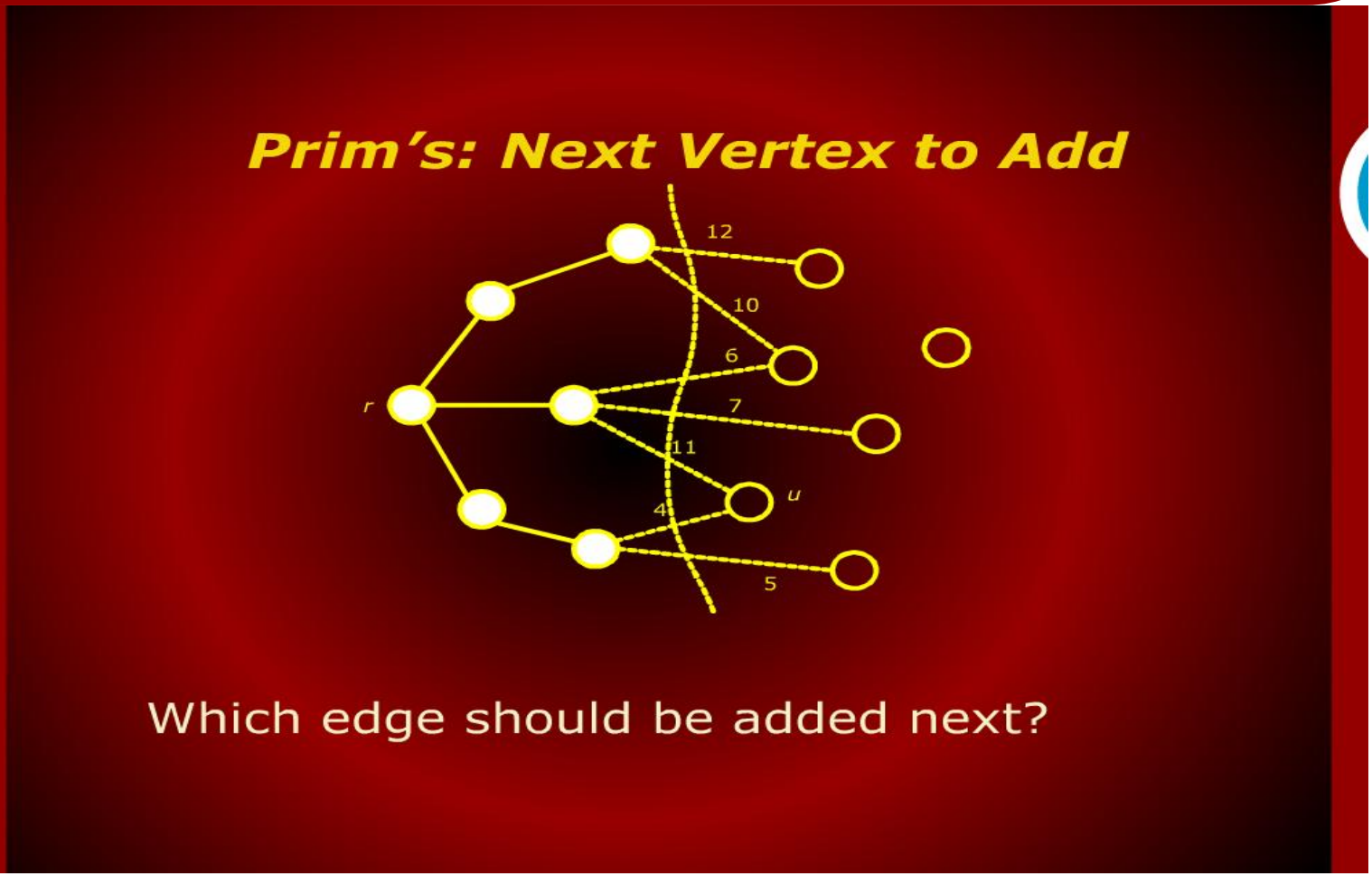
Prime's Next Vertex to Add



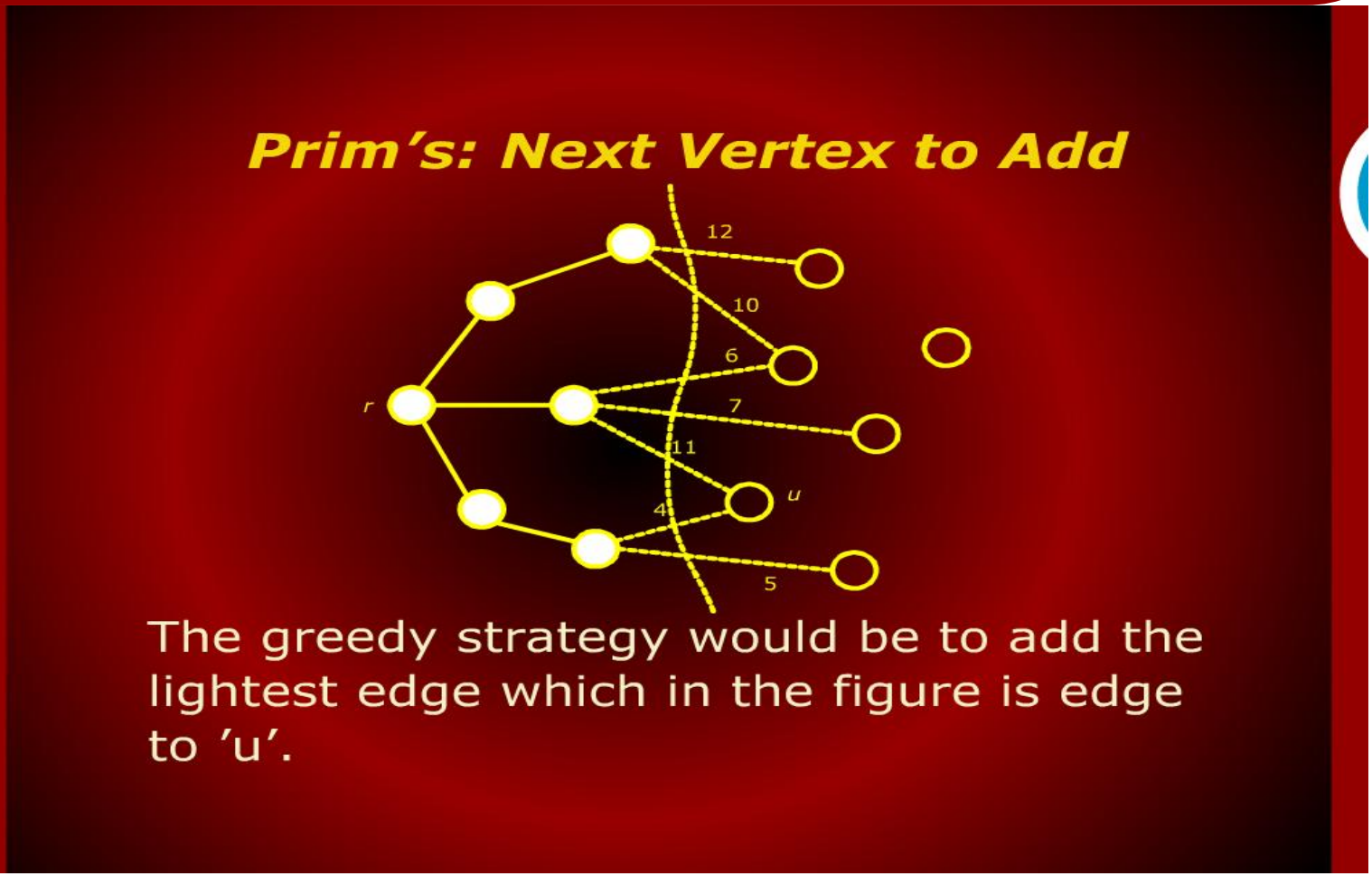
Prime's Next Vertex to Add



Prime's Next Vertex to Add



Prime's Next Vertex to Add



Shortest Path Variants

Shortest Path Variants

Shortest Path Variants

Prim's Algorithm: Priority Queue

Prim's Algorithm: Priority Queue

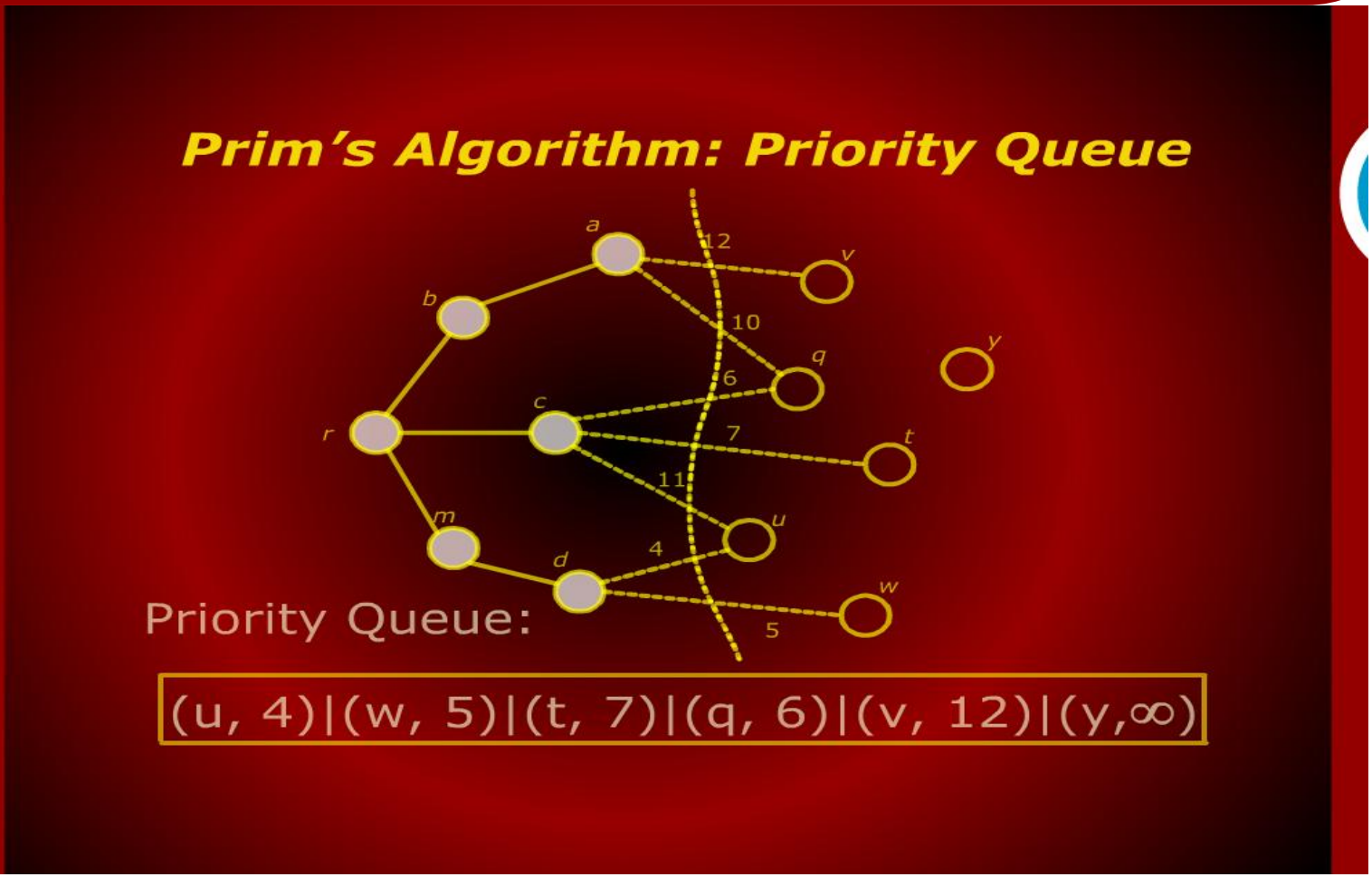
- What do we store in the priority queue?
- It may seem logical that edges that cross the cut should be stored since we choose light edges from these.
- Although possible, there is more elegant solution which leads to a simpler algorithm.

Prim's Algorithm: Priority Queue

Prim's Algorithm: Priority Queue

- For each vertex u $\in$ (V - S) (not part of the current spanning tree), we associate a key key[u].
- The key[u] is the weight of the lightest edge going from u to any vertex in S.
- If there is no edge from u to a vertex in S, we set the key value to ∞.
- We also store in pred[u] the end vertex of this edge in S.

Prim's Algorithm: Priority Queue

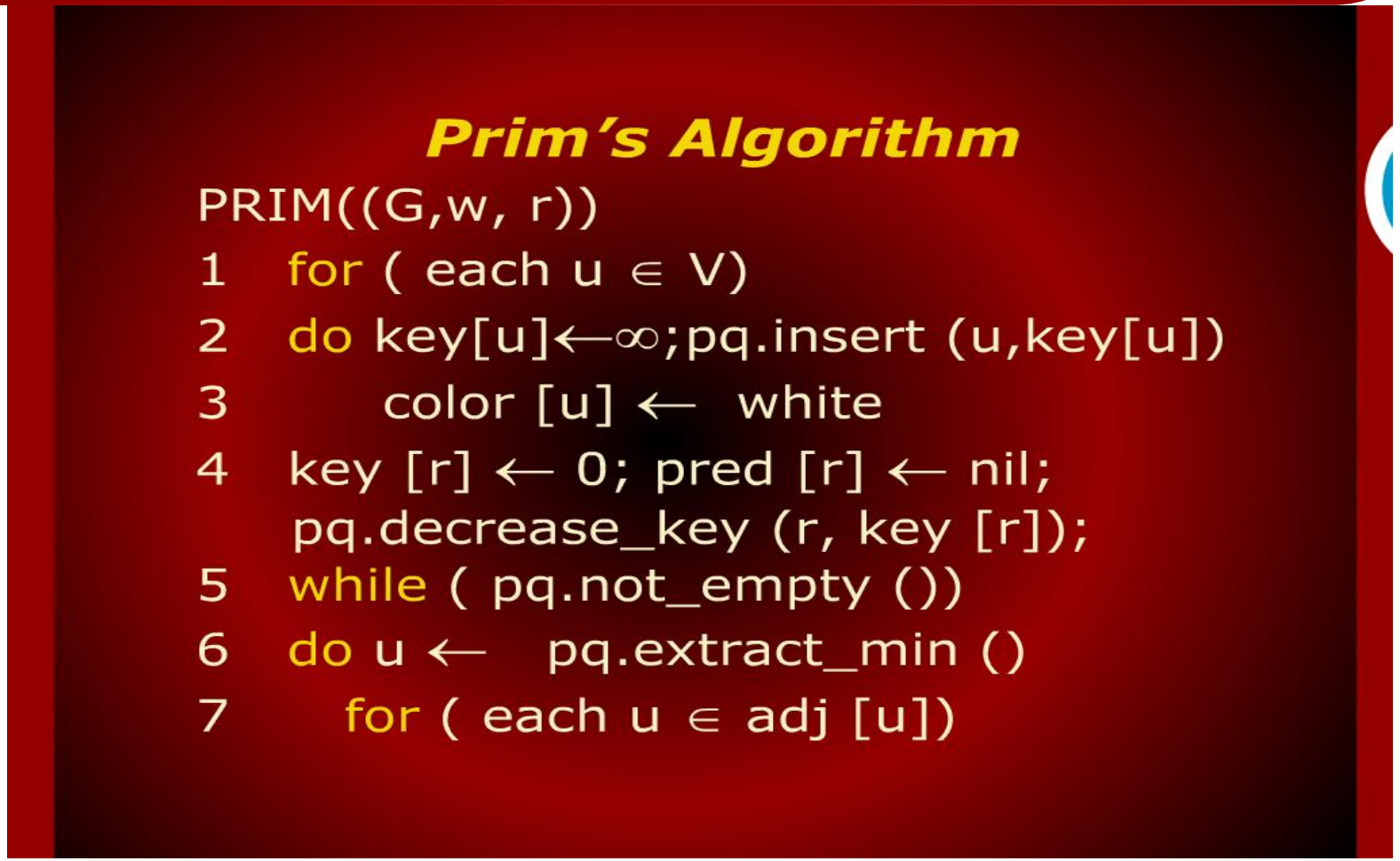


Prim's Algorithm

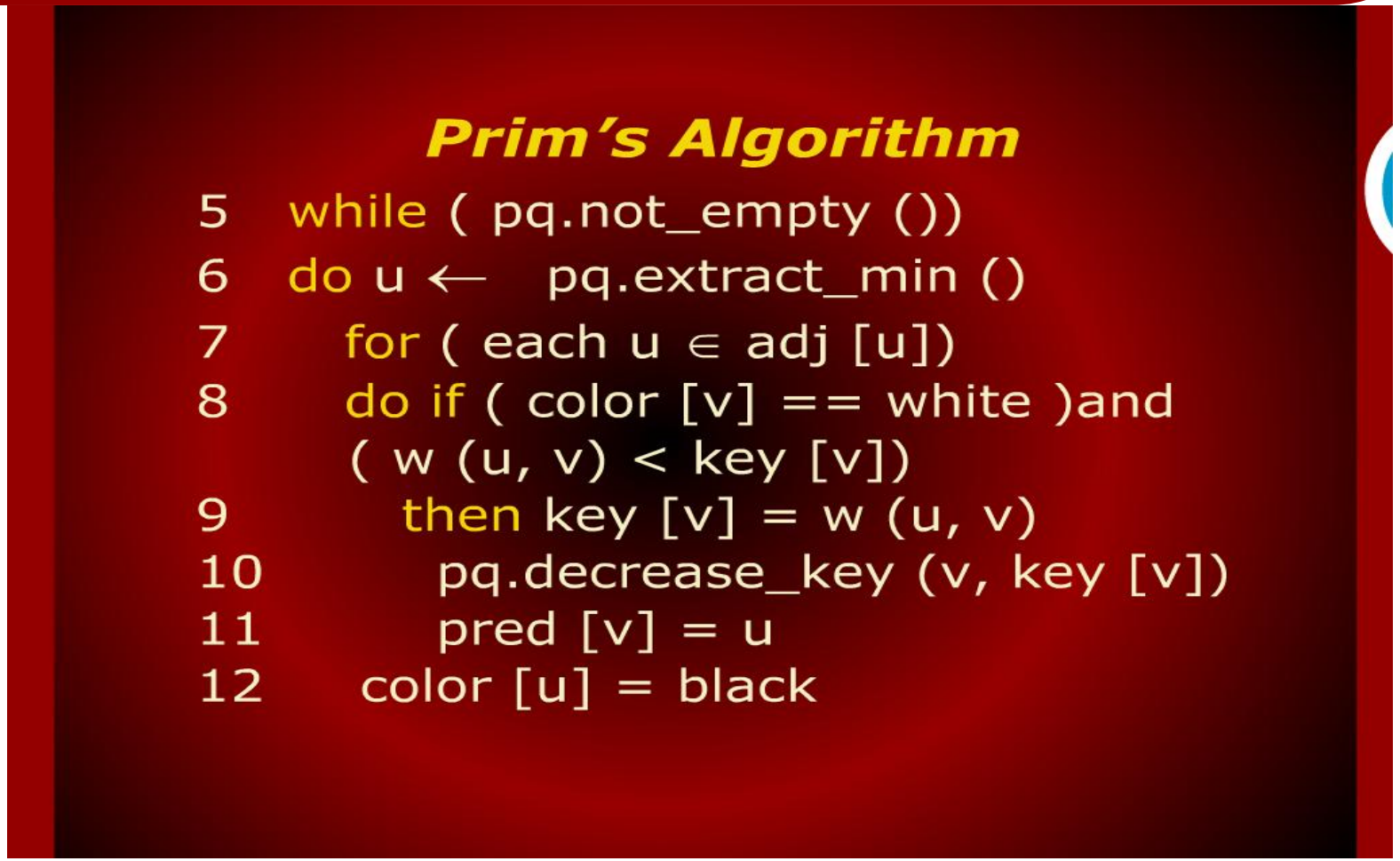
Prim's Algorithm

- We will also need to know which vertices are in S and which are not.
- To do this, we will assign a color to each vertex.
- If the color of a vertex is black then it is in S otherwise not.

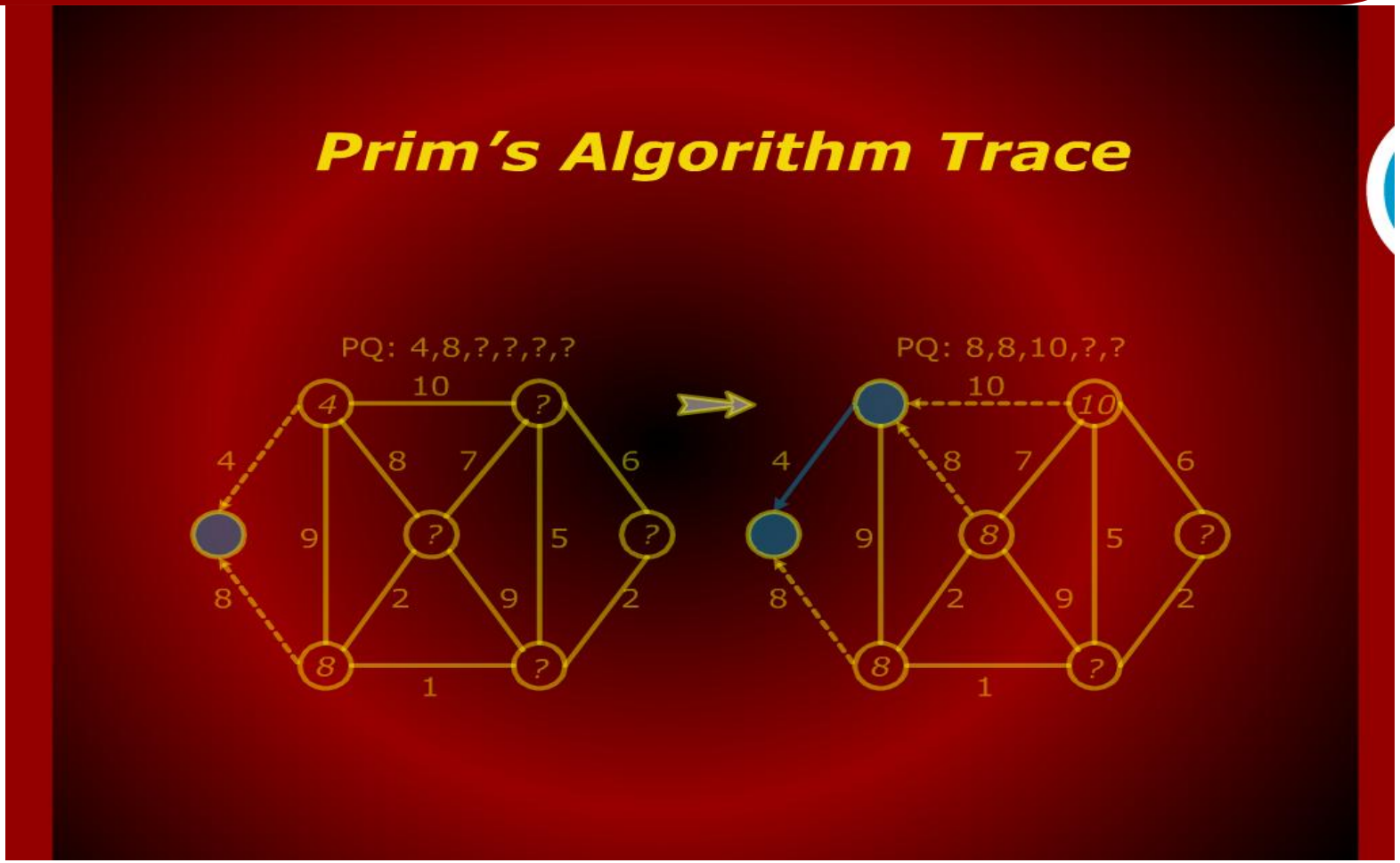
Prim's Algorithm



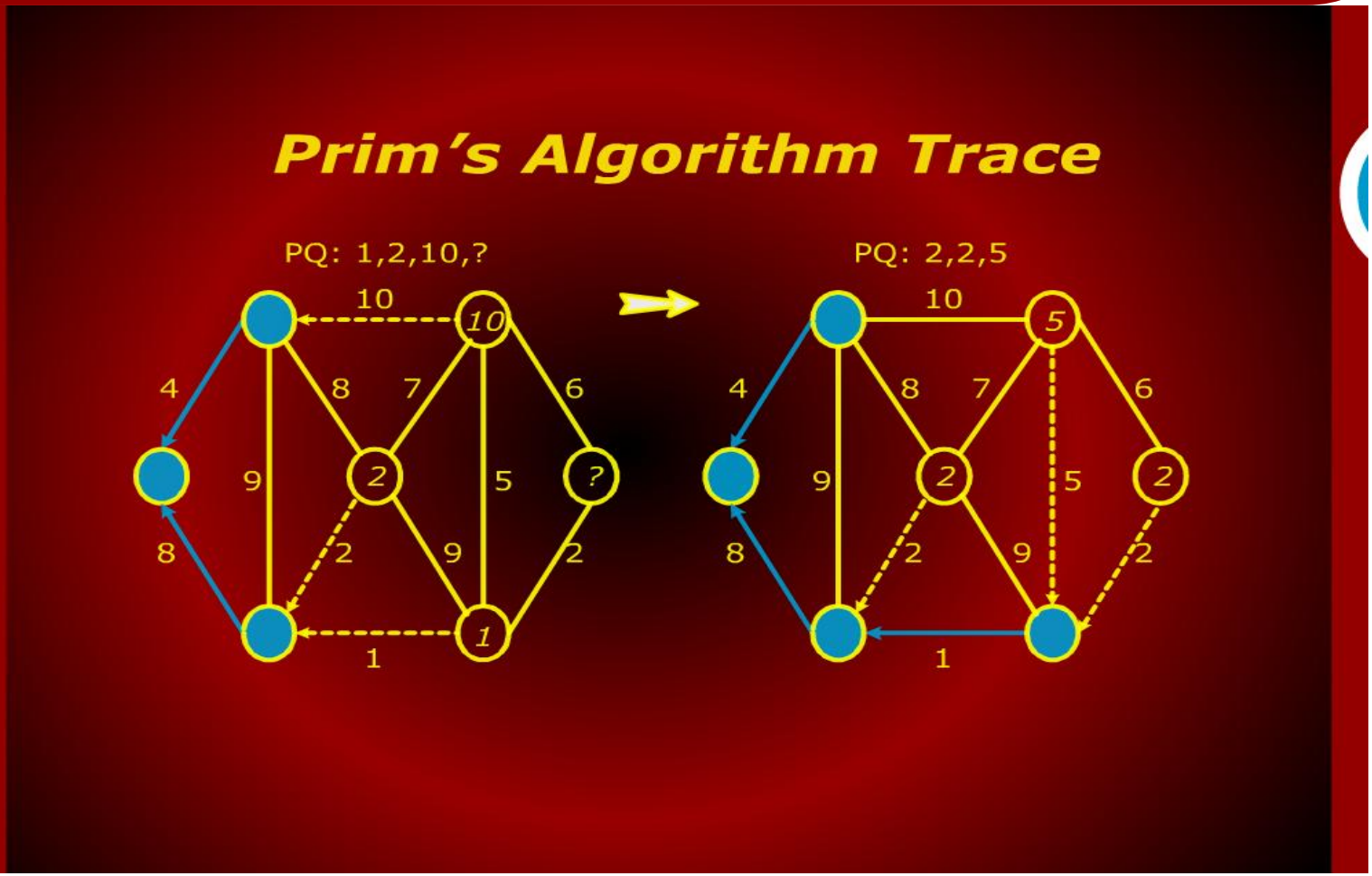
Prim's Algorithm



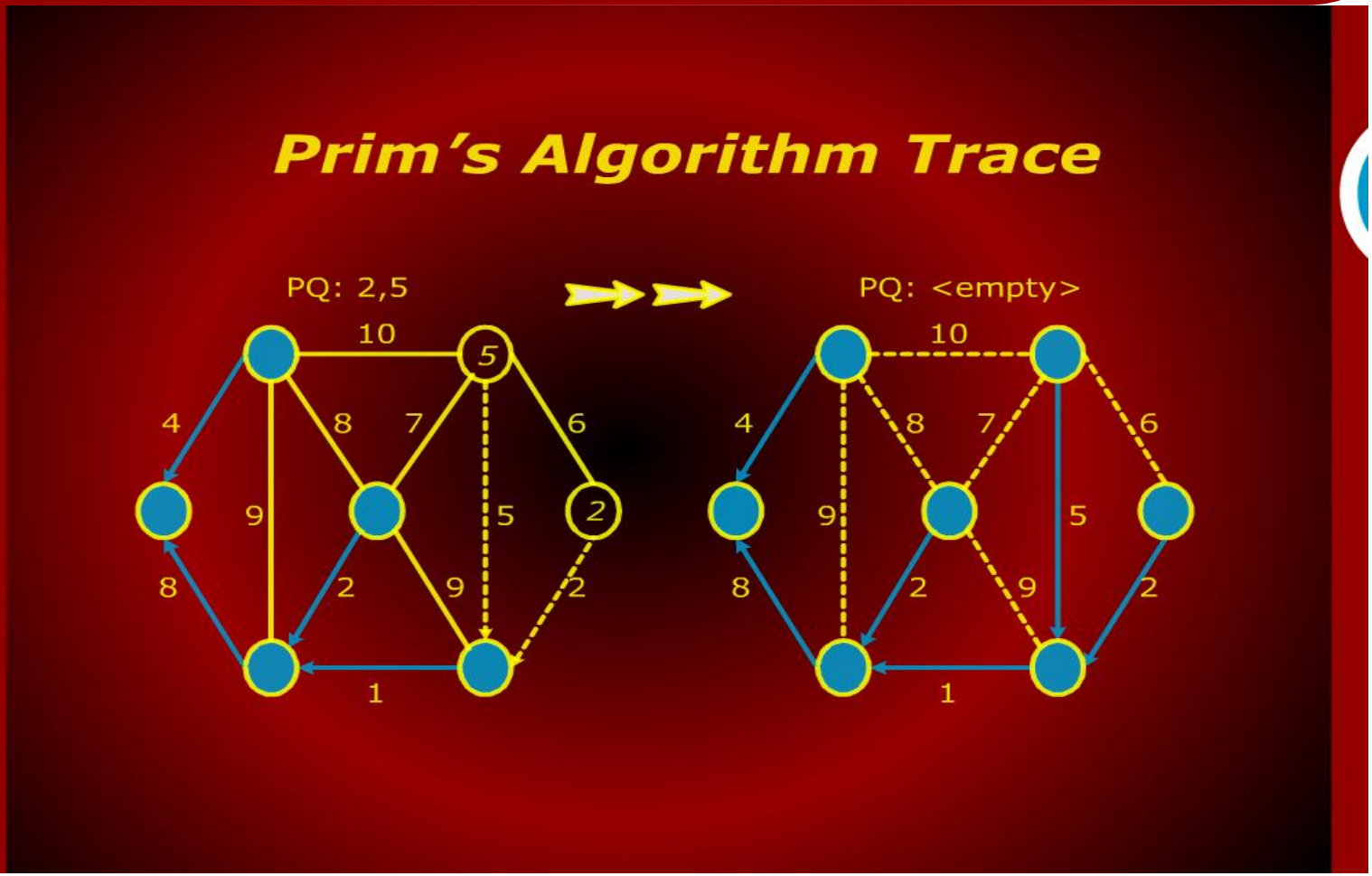
Prim's Algorithm Trace



Prim's Algorithm Trace



Prim's Algorithm Trace



Prim's Algorithm Run Time Analysis

Prim's Algorithm Runtime Analysis

- It takes O(log V) to extract a vertex from the priority queue.
- For each incident edge, we spend potentially O(log V) time decreasing the key of the neighboring vertex.
- Thus the total time is O(log V + deg(u) log V).
- The other steps of update are constant time.

Prim's Algorithm Run Time Analysis

Prim's Algorithm Runtime Analysis

So the overall running time is

$$T(V, E) = \sum_{u \in V} (\log V + \deg(u) \log V)$$

$$= \log V \sum_{u \in V} (1 + \deg(u))$$

$$= (\log V)(V + 2E)$$

$$= \Theta((V + E) \log V)$$

Since G is connected, V is asymptotically

Prim's Algorithm Run Time Analysis

Prim's Algorithm Runtime Analysis

$$= \log V \sum_{u \in V} (1 + \deg(u))$$

$$= (\log V)(V + 2E)$$

$$= \Theta((V + E) \log V)$$

Since G is connected, V is asymptotically no greater than E so this is $\Theta(E \log V)$, same as Kruskal's algorithm.

Shortest Path

Shortest Path

Shortest Path

- A motorist wishes to find the shortest possible route between Peshawar and Karachi.
- Given a road map of Pakistan on which the distance between each pair of adjacent cities is marked
- Can the motorist determine the shortest route?

Prim’s Algorithm